



PRODUCT REQUIREMENTS DOCUMENT (PRD)

TEAM NAME: 4CODEHERS

Challenge: Scenario 1: Global Stablecoin Payment Relay

TEAM MEMBERS & ROLES: ALONDRA, FELIPA, GELSI, AND SAMARA

Name of App: GlöbaLink

Problem Statement

Issue solving

statement:

- Millions of people around the world—including students, migrant workers, and small business owners—struggle to send or receive money internationally due to high fees, long delays, and lack of access to traditional banks. Cross-border remittances for unbanked or underbanked populations remain slow, expensive, and inaccessible.

Who's affected?

- Those who do not have a bank account and foreign students living in rural communities with low bandwidth and immigrants.

The importance of this solution for the future of payments:

- The world is becoming larger with a need for connectivity; getting into smaller pockets is key!

Example (Stablecoin Challenge): Cross-border remittances to rural areas remain slow and expensive, limiting financial inclusion.

Example (AI Challenge): Customers face frequent false fraud alerts, causing frustration and distrust in digital banking.

Objective/Solution Overview

30-second elevator pitch:

GlöbaLink is redefining financial access to rural areas. Our solar-powered ATM kiosks let people exchange and send money instantly, even without internet or electricity. With biometric security, transparent fees, and seamless wallet integration, GlöbaLink empowers communities to connect to the global economy — safely, affordably, and sustainably.

User Personas & Journey

STEPHANY ALEMAN

- **Who she is:** A Mom from rural Mexico.
- **Pain Points:**
 - Must travel to bank for Western Union.
 - Doesn't receive the full amount (\$500 reduced by fees).
 - Waits days for funds; bank hours are limited.
- **Goals:** Wants faster, local access to money at fair rates.
- **User Journey Example:**
 - **Step:** Son sends transfer.
 - **Action:** Stephany uses mobile/Wi-Fi or a low-bandwidth app in her village.
 - **System Response:** Funds received instantly, no extra travel.
 - **Pain Point Solved:** Convenience + lower cost.

ABNER LA PIERRE

- **Who he is:** Foreign exchange student in the U.S.
- **Pain Points:**
 - No U.S. bank account.
 - Relies on credit card → pays high foreign transaction fees.
 - Trouble storing/receiving cash from home.
- **Goals:** Needs an easy way to receive/send money digitally, without fees.
- **User Journey Example:**
 - **Step:** Parent sends funds internationally.
 - **Action:** Abner receives instantly via mobile app wallet.
 - **System Response:** Funds available, no foreign transaction charges.
 - **Pain Point Solved:** Faster, cheaper access to money.

GABRIELLE SOLIS

- **Who she is:** First-generation American, supports parents and kids.
- **Pain Points:**
 - Sending money is slow and expensive.
 - Parents only get a partial amount after fees.
 - Hassle of arranging pickup for her parents.
- **Goals:** Wants reliable, low-cost, direct transfer to parents' phones.
- **User Journey Example:**
 - **Step:** Gabrielle initiates transfer.
 - **Action:** Parents get funds directly on their mobile devices.
 - **System Response:** Full amount received quickly, with clear fees.
 - **Pain Point Solved:** Saves time, reduces cost, and provides peace of mind.

Core Features & Functionality

5 essential features of your MVP

1. Resilient Transaction Connectivity

- The kiosk must process money exchange and transfer transactions using a **SIM card and satellite connectivity**, ensuring continuous operation without reliance on traditional internet or broadband infrastructure.
- The system should queue transactions when the connection is temporarily weak and automatically sync once signal strength stabilizes.

2. Biometric User Authentication

- Each kiosk must integrate fingerprint or facial recognition for secure and seamless user verification.
- This eliminates the need for physical cards or PINs and reduces fraud risk.

3. Multi-Currency Exchange Module

- The platform must support secure currency exchange between local and international currencies, including dynamic rate updates through the satellite/SIM network.
- Conversion logic should handle both digital and fiat transactions.

4. Solar-Powered Kiosk Infrastructure

- Each kiosk must operate fully on solar energy with a battery backup, ensuring 24/7 uptime in rural and off-grid areas.
- Components must be durable, weatherproof, and energy efficient.

5. Mobile Wallet and Banking Integration

- The system must allow users to link their mobile wallets or bank accounts for deposits, withdrawals, and remittances.
- APIs should ensure interoperability with financial institutions, digital wallets, and mobile money platforms.

Compliance & Security

1. Regulatory Frameworks

- a. GlobalLink follows major global compliance standards, including
 - i. **MiCA (Markets in Crypto-Assets Regulation)**—ensures stablecoin transparency, consumer protection, and accountability in the EU.
 - ii. **GDPR (General Data Protection Regulation)**—safeguards data privacy and gives users full control over their personal information.
 - iii. **CCPA (California Consumer Privacy Act)**—protects U.S. users' data rights and transparency.
 - iv. **FinCEN & FATF Guidelines**—align with international anti-money laundering (AML) and counter-terrorism financing (CTF) standards.

2. AML/KYC & Fraud Detection

- a. Use **AI-based identity verification** and **risk scoring** to detect suspicious activity in real time.
- b. Integrates **KYC (Know Your Customer)** checks using light verification for accessibility while meeting compliance requirements.
- c. Tracks and flags high-risk transactions automatically through smart contract monitoring.

3. Data Security & Encryption

- a. All transactions are secured with **end-to-end encryption** and **blockchain immutability** to prevent tampering.
- b. Employs **zero-knowledge proof** to verify user identities without revealing private data.

4. User Transparency & Trust

- a. Users can view their compliance status and transaction history at any time.
- b. Clear privacy policy and permission-based data sharing ensure users know exactly how their data is handled.

Business Model & Impact

Who Pays / Who Benefits

- **Users:** Students, migrant workers, freelancers, and small businesses benefit from lower fees and faster payments.
- **Partners:** Banks, fintech companies, and mobile wallets integrate GlobalLink's API for cross-border transfers, WhatsApp.
- **Businesses:** Save time and money through instant global settlements and transparent exchange rates.

Revenue Path

- **Transaction Fee:** 1–2% per transfer, significantly lower than traditional remittance services
- **Currency Conversion Spread:** 1-2% fee per conversion from stablecoin-to-local-currency conversions.

Social & Economic Impact

- Reduces global remittance fees by up to **70%**, putting more money directly into users' hands.
- Boosts **financial inclusion** for unbanked populations with simple ID verification and mobile access.
- Speeds up **international trade and family remittances**, stimulating local economies.
- Promotes **trust and transparency** in global finance using blockchain technology even in underdeveloped communities.

- Increased network capacity through mobile accessibility in areas of low bandwidth, especially expanded through our Kiosk infrastructure

Tech Stack & Tools

APIs & Payment Gateways

- **SWIFT API (International Transfers):**
 - When a user wants to send money abroad, the kiosk connects to the SWIFT network via its API.
 - The system securely sends transaction details (amount, recipient bank, and account info) to the SWIFT network.
 - SWIFT processes the transfer between banks, and GlöbaLink receives confirmation once completed.
 - This ensures fast, reliable international transfers while maintaining compliance with banking standards.
- **SEPA API (Euro-Denominated Transfers):**
 - For users sending or receiving euros within SEPA countries, the kiosk connects to the SEPA network.
 - Transaction instructions (amount, IBAN, recipient bank) are sent via the SEPA API.
 - The SEPA system settles the transfer, and GlöbaLink updates the user instantly with transaction status.
 - This allows fast, low-cost transfers across Europe with standardized processing.
- **WhatsApp wallet Integration:** Allows users to send or receive payments directly through familiar apps.

Data Sources:

- **Blockchain data:** Real-time transaction records from Ethereum, Polygon, and Solana networks.
- **Exchange rate APIs:** Live crypto-to-fiat conversion data (e.g., CoinGecko, Chainlink Oracles).
- **Compliance & KYC APIs:** User identity verification and AML screening data from trusted providers (e.g., Sumsb, Trulioo).

Languages:

- **Solidity**—for writing smart contracts on Ethereum.
- **JavaScript / TypeScript**—for front-end and back-end logic.
- **Python**—for AI-based fraud detection and data analytics.

Frameworks & Tools:

- **React.js + Tailwind CSS** – for building a modern, responsive front-end UI.
- **Node.js + Express.js** – for server-side operations and API management.
- **MongoDB**—for secure user and transaction data storage.
- **Vibes.diy**—for UI wireframes, prototype visualization, and dashboard design.

Vibes.diy for Prototype Development

- a. Use **Vibes.diy** to design interactive wireframes, user dashboards, and visual transaction flows.
- b. Generate UI mockups of the **wallet interface, transfer screen, and analytics dashboard** directly from our PRD (Product Requirement Document).

Example: We'll use vibes.diy to generate front-end wireframes and visual data dashboards using this PRD prompt.

Success Metrics

How will you measure success?

Metric – Target – Why It Matters]

Adoption and Growth (Early Stages)

Metric	Target	Why It Matters
Active Wallets	54K+ in Year 1	Proves product market fit and user traction.
Monthly Active Users (MAU)	≥60% of wallets	Measures repeat use and retention.
Referral Conversion Rate	25% of new users	Demonstrates trust-based, organic growth.

Trust and Transparency (Measures resilience)

Metric	Target	Why It Matters
Reserve Ratio (Proof-of-Reserves)	100% backed, verified	Builds user and regulator confidence.
Community Verifier Nodes	100+ in pilot year	Shows decentralization and shared oversight.
Attestation Frequency	Weekly → Real-time (by Year 2)	Reflects transparency and system maturity.

Measures Regulatory Resilience

Metric	How to Measure	Why It Matters
Regulatory Compliance Rate	% of audits passed without critical findings	Prevents shutdowns or fines.
Time to Implement Compliance Updates	Average days to adjust for new regulations	Indicates agility under external changes.
Partner/Custodian Reliability	% of partners meeting SLA for reserves or KYC	Ensures ecosystem reliability and avoids single points of failure.

Prototype Plan (Prompt Starter)

Turn your PRD into a prompt for Vibes.diy:

Example: “Create a working prototype for a cross-border stablecoin remittance app that can work offline using SMS, includes dynamic KYC based on the size of the transaction, and provides a transparent fee dashboard. Include UI screens for sender, receiver, and compliance admin.”]

Presentation Snapshot

[Prepare a 3-5 minute pitch using the provided PowerPoint template covering:

1. The problem that is solved
2. Why it matters
3. The key features of their MVP

4. How they used AI or Vibes.diy to accelerate innovation
5. The real-world impact or potential to go-to-market]